SUPPLY CHAIN DELIVERY DELAY PREDICTION

Machine Learning Project Report

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Tools Used: Python, Pandas, Scikit-learn, Matplotlib, Seaborn, XGBoost

1. Introduction

In modern supply chain systems, timely delivery is a critical factor influencing customer satisfaction and operational efficiency. However, delays in shipment can occur due to various factors such as logistics bottlenecks, shipping schedules, high order volumes, and operational inefficiencies.

Predicting delivery delays in advance allows organizations to proactively manage logistics operations, optimize scheduling, and minimize customer dissatisfaction.

This project focuses on building a **machine learning model that predicts whether a delivery will be late** using historical supply chain data. By analyzing order characteristics, shipping details, and product information, the model can identify patterns associated with delivery delays.

Such predictive systems can help logistics companies improve operational decision-making and ensure better delivery performance.

2. Problem Statement

Delivery delays are a common challenge in supply chain management. Companies need tools that can anticipate potential delays before they occur.

The goal of this project is to develop a **machine learning model capable of predicting late delivery risk** based on supply chain transaction data.

The model will analyze various features such as:

- shipping mode
- order quantity
- product price
- order region
- customer segment
- order date information

Using these features, the model predicts whether a shipment will be delivered late.

3. Dataset Description

The dataset used in this project is the **DataCo Smart Supply Chain Dataset**, obtained from Kaggle.

This dataset contains detailed records of supply chain transactions including information about customers, orders, shipping modes, product details, and logistics operations.

Dataset Characteristics

Attribute	Description
Total Records	180,519
Original Features	50+
Target Variable	Late_delivery_risk
Data Type	Mixed (categorical and numerical)

Target Variable

The target variable used for prediction is:

Late_delivery_risk

Value	Meaning
0	Delivery on time
1	Delivery late

This makes the problem a **binary classification problem**.

4. Data Cleaning

Raw datasets often contain missing values, redundant columns, and irrelevant features that can negatively impact model performance.

Several preprocessing steps were performed to clean the dataset.

Handling Missing Values

Missing values were identified using:

df.isnull().sum()

Numerical columns with missing values were filled using **median imputation**, which helps preserve the statistical distribution of the data.

Removing Irrelevant Columns

Some columns were removed because they were not useful for prediction or contained redundant information. Examples include:

- Customer Email

- Customer Password
- Product Image
- Product Description

Removing such columns reduces noise in the dataset and improves model training efficiency.

Handling Target Leakage

Columns such as **Days for shipping (real)** and **Days for shipment (scheduled)** were removed because they directly reveal whether a delivery is late. Including these variables would cause the model to cheat and produce unrealistic results.

5. Feature Engineering

Feature engineering is an important step in machine learning that transforms raw data into useful features for model training.

Date Feature Extraction

The dataset contained order dates that were converted into more useful features:

- **order_day**
- **order_month**

These features allow the model to capture temporal patterns in deliveries.

Encoding Categorical Variables

Machine learning algorithms require numerical inputs. Therefore categorical variables such as:

- Shipping Mode
- Customer Segment
- Market
- Order Region

were converted into numerical form using **One-Hot Encoding**.

Example:

df = pd.get_dummies(df, drop_first=True)

This transformation allows the model to learn relationships between categorical features and delivery delays.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) helps understand patterns and relationships in the dataset.

Several visualizations were created using **Matplotlib** and **Seaborn**.

Late Delivery Distribution

This plot shows the number of deliveries that were late compared to those delivered on time. It helps identify whether the dataset is balanced.

Shipping Mode Analysis

Shipping modes such as Same Day, Second Class, and Standard Class were analyzed to observe how shipping methods influence delivery delays.

Correlation Heatmap

A correlation heatmap was generated to identify relationships between numerical variables and detect important features affecting delivery delays.

EDA helped reveal patterns in the data and guided feature selection for model training.

7. Machine Learning Models

Two machine learning models were trained and evaluated in this project:

1. Random Forest Classifier
2. XGBoost Classifier

Random Forest

Random Forest is an ensemble learning method that builds multiple decision trees and combines their predictions. It is effective for handling complex datasets and capturing non-linear relationships.

XGBoost

XGBoost is a gradient boosting algorithm that sequentially builds decision trees and improves performance by correcting previous errors.

Both models were trained using the **training dataset** and evaluated on the **test dataset**.

8. Model Training

The dataset was split into training and testing sets using the following ratio:

- Training Data: 80%
- Testing Data: 20%

Example code:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
)
```

This ensures that the model is evaluated on unseen data.

9. Model Evaluation

Model performance was evaluated using **Accuracy**.

Accuracy Formula

Accuracy represents the proportion of correct predictions made by the model.

Accuracy = Correct Predictions / Total Predictions

Model Performance

Model	Accuracy
Random Forest	74.6%
XGBoost	73.2%

Random Forest achieved slightly higher accuracy and was therefore selected as the final model.

10. Learning Curve Analysis

A learning curve was generated to evaluate model performance as the size of the training dataset increases.

The learning curve compares:

- Training accuracy
- Validation accuracy

The training accuracy remained close to 1.0 while validation accuracy stabilized around 0.64, indicating slight overfitting. However, the model still generalized reasonably well with a test accuracy of approximately 74.6%.

11. Model Comparison Visualization

A bar chart was created to compare the performance of Random Forest and XGBoost models.

The visualization clearly shows that **Random Forest slightly outperforms XGBoost**, making it the better choice for this dataset.

12. Conclusion

This project demonstrates how machine learning techniques can be applied to supply chain data to predict delivery delays.

By performing data cleaning, feature engineering, exploratory analysis, and model training, a predictive model was developed that can identify shipments likely to be delayed.

Among the models tested, **Random Forest achieved the best performance with an accuracy of approximately 74.6%**.

Such predictive systems can assist logistics companies in:

- identifying high-risk deliveries
- improving shipping strategies
- optimizing supply chain operations
- enhancing customer satisfaction

Future improvements could include hyperparameter tuning, advanced feature engineering, and incorporating real-time logistics data.